

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 9

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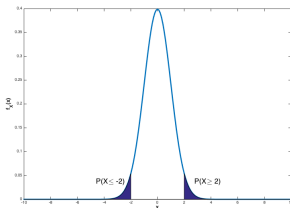
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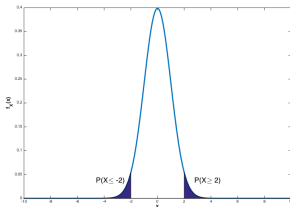
Normal Distribution

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 - Stock returns.
 - Molecular velocities.
 - Locations of projectiles aimed at a target.



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- Further, it has a number of nice properties that make it easy to work with. Like symmetry. In the above picture, $P(X \geq 2) = P(X \leq -2)$.

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- What are the mean and variance of Y ?
- $\mathbb{E}[Y] = a\mu + b$
- $\text{Var}[Y] = a^2\sigma^2$.
- In fact, if $Y = aX + b$, then Y is also a normal random variable, with mean $a\mu + b$ and variance $a^2\sigma^2$. Thus, $Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.
We will see the proof of it when we will discuss distributions of functions of random variable.

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- So, $P(Y < 0) = P(Z < -1) = P(X < -1) = 0.159$.

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- So, $y/2 = -1$ and as a result $y = -2$!

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- If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$.
- (Note, we often use the letter Z for standard normal random variables)

Standard Normal Distribution

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z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6951	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
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- $P(Z < 0.21) = 0.5832$

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- And we’re done! $P(X < 0) = 1 - \Phi(1.5)$
- Look at the table. $P(X < 0) = 1 - 0.9332 = 0.0668$

Exponential Random Variable

- How to model the amount of time until something happens, such as
- the next email arrives
- an accident happens
- a light bulb burns out
- Notation: $X \sim \text{Exp}(\lambda)$

The exponential distribution

- An exponential r.v. has pdf and cdf:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$F_X(x) = \int_0^x \lambda e^{-\lambda x} dx = 1 - e^{-\lambda x}$$

- $\mathbb{E}[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = \frac{1}{\lambda} \int_0^\infty (\lambda x) e^{-\lambda x} d(\lambda x) = \frac{1}{\lambda} \int_0^\infty u e^{-u} du = \frac{1}{\lambda}$
- $\text{var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \int_0^\infty x^2 \lambda e^{-\lambda x} dx - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \int_0^\infty u^2 e^{-u} du - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$

Multiple Random Variables

- We can also have multiple continuous random variables associated with the same experiment/sample space.
- For example, our experiment might be selecting a randomly selected person.
- The sample space would be the set of all possible characteristics of this person.
- We can summarize these characteristics into continuous random variables, e.g.
 - The person's height
 - The person's weight
 - The person's age
- Again, multiple random variables stemming from the same sample space!
- These random variables will often depend on each other: Knowing a person is taller than 6'5" tells us something about their expected weight.

Multiple Continuous Random Variables

- Let X and Y be two continuous random variables.
- Each one takes on values on the real line, i.e. $X \in \mathbb{R}$ and $Y \in \mathbb{R}$.
- Together, each possible pair of values describe a point in the real plane, i.e. $(X, Y) \in \mathbb{R}^2$.
- We say X and Y are jointly continuous if the probability of them jointly taking on values in some subset B of the plane can be described as

$$P((X, Y) \in B) = \int_{(x,y) \in B} f_{X,Y}(x, y) dx dy$$

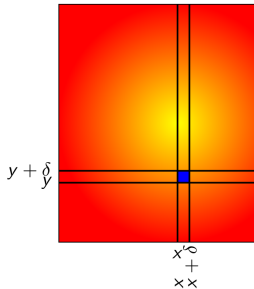
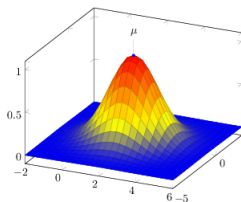
using some continuous function $f_{X,Y}$, for all $B \in \mathbb{R}^2$ - i.e. all subsets of the 2-D plane.

- Notation means “integrate over all values of x and y s.t. $(x, y) \in B$.”

- We call $f_{X,Y}$ the joint pdf of X and Y .
- It allows us to calculate the probability of any set of combinations of X and Y .
 - e.g. the probability that a person weighs over 200lb and is under 6'
 - e.g. the probability that a person's height in inches is more than twice their weight in pounds.
 - So, this could describe the first scenario above,
 $P(200 \leq X < \infty, -\infty < Y \leq 6)$
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Joint PDF Intuition



- We can think of the joint PDF $f_{X,Y}(x,y)$ as the “probability mass per unit area” for a small area near X .
- Again, remember, $f_{X,Y}(x,y)$ is not a probability!

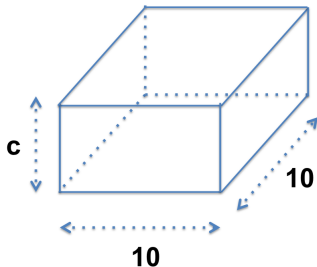
Multiple Random Variables to Single Random Variable

- We can get from the joint PDF of X and Y to the marginal PDF of X by integrating over (marginalizing over) Y:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

Bivariate Uniform Random Variable

- Anita (X) and Benjamin (Y) both pick a number between 0 and 10, according to a continuous uniform distribution. What is $f_{X,Y}(x,y)$?



- Let's see... we know all pairs (x,y) are equally likely, so we know $f_{X,Y} = c$. It must satisfy

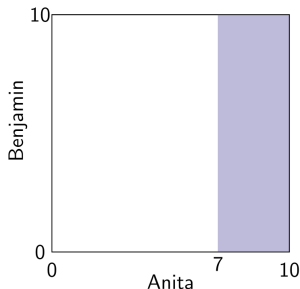
$$\int_0^{10} \int_0^{10} f_{X,Y}(x,y) dx dy = 1$$

- So, $c = f_{X,Y}(x,y) = 0.01$ for all $0 \leq x,y \leq 10$.

- $f_{X,Y}(x,y) = \begin{cases} 0.01, & \text{If } x,y \in [0,10] \\ 0, & \text{otherwise} \end{cases}$
- What is $f_X(x)$?
- $f_X(x) = \int_0^{10} 0.01 \, dy = \begin{cases} 0.1, & \text{If } x \in [0,10] \\ 0, & \text{otherwise} \end{cases}$
- Not surprisingly $X \sim \text{Uniform}([0,10])$ and $Y \sim \text{Uniform}([0,10])$.
- In general, we will have $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy$
- We call $f_X(x)$ the marginal PDF of X .

Example: Marginalization

- What is the probability that Anita picks a number greater than 7?



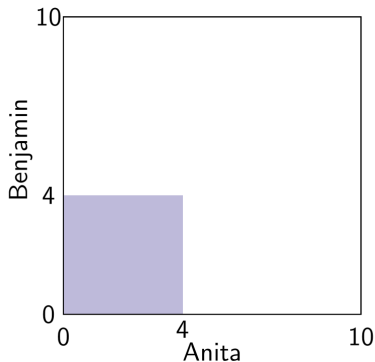
$$P(X > 7) = \int_{x=7}^{10} \int_{y=0}^{10} 0.01 dx dy = 0.3$$

- That's going to correspond to the shaded region...
 $P(X > 7) = 0.01(3 \times 10) = 0.3.$
- That does not depend on Benjamin. Thus, it is same as

$$P(X > 7) = \int_7^{10} f_X(x) dx = \int_7^{10} 0.1 dx = 0.3$$

Example Uniform Random Variable

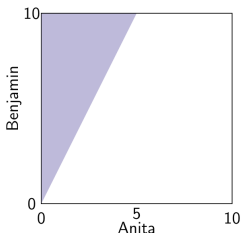
- What is the probability that they both pick numbers less than 4?



- It will be $P(X < 4, Y < 4) = \int_0^4 \int_0^4 0.01 \, dx \, dy = 0.16$
- i.e., $0.01 \times \text{the shaded area} = 16/100!$

Example: Uniform Random Variable

- What is the probability that Benjamin picks a number at least twice that of Anita?



- Its the area of the shaded region. $P(Y \geq 2X) = 0.01\left(\frac{5 \times 10}{2}\right) = 0.25$.
- Or, using calculus:

$$\begin{aligned} P(Y \geq 2X) &= \int_{x=0}^{10} \int_{y=2x}^{10} f_{X,Y}(x,y) dx dy \\ &= 0.01 \int_{x=0}^{10} \int_{y=2x}^{10} \mathbb{I}_{[0 \leq x \leq 10, 0 \leq y \leq 10]} dx dy = 0.01 \int_{x=0}^{10} \int_{y=2x}^{10} \mathbb{I}_{[0 \leq 2x \leq 10]} dx dy \\ &= 0.01 \int_{x=0}^5 (10 - 2x) dx = 0.01(10 \times 5 - 5^2) = 0.01 \times 25 = 0.25 \end{aligned}$$

- Recall, the PDF of a continuous random variable X is the non-negative function $f_X(x)$ that satisfies

$$P(X \in B) = \int_{x \in B} f_X(x) dx$$

for any subset B of the real line.

- Let A be some event with $P(A) > 0$
- The conditional PDF of X , given A , is the non-negative function $f_{X|A}$ that satisfies $P(X \in B | X \in A) = \int_{x \in B} f_{X|A}(x) dx$ for any subset B of the real line.
- If B is the entire line, then we have $\int_{-\infty}^{\infty} f_{X|A}(x) dx = 1$
- So, $f_{X|A}(x)$ is a valid PDF.

- The event we are conditioning on can also correspond to a range of values of our continuous random variable.

- **Definition**

$$f_{X|\{X \in A\}}(x) = \begin{cases} \frac{f_X(x)}{P(X \in A)}, & \text{if } X \in A \\ 0, & \text{otherwise} \end{cases}$$

- In this case, we can write the conditional probability as

$$\begin{aligned} P(X \in B | X \in A) &= \int_{x \in B} f_{X|\{X \in A\}} dx = \int_{x \in B} \frac{f_X(x) \mathbb{I}_{[X \in A]}}{P(X \in A)} dx \\ &= \int_{A \cap B} \frac{f_X(x)}{P(X \in A)} dx = \frac{P(\{X \in A\} \cap \{X \in B\})}{P(X \in A)} \end{aligned}$$

- This is a valid PDF—non-negative and integrates to one.

Conditioning: Memoryless property of exponential distribution

- $X \sim \text{Exp}(\lambda)$
- $P(X \geq s + t | X \geq s) = ?$
- Remember the exponential? $F_X(x) = 1 - e^{-\lambda x}$.
- $$P(X > s + t | X > s) = \frac{P(X > s + t, X > s)}{P(X > s)} = \frac{P(X > s + t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X > t)$$

Conditioning: Memoryless property of exponential distribution

- $X \sim \text{Exp}(\lambda)$
- $f_{X|X>s}(x) = \frac{\lambda e^{-\lambda x}}{P(X>s)} = \begin{cases} \lambda e^{\lambda(x-s)}, & \text{If } x > s \\ 0, & \text{Otherwise} \end{cases}$
- $P(X > s + t | X > s) = ?$
- Remember the exponential? $F_X(x) = 1 - e^{-\lambda x}$.
- $P(X > s + t | X > s) = \int_{s+t}^{\infty} f_{X|X>s}(x) dx = \lambda \int_{s+t}^{\infty} e^{-\lambda(x-s)} ds = \lambda \int_t^{\infty} e^{-\lambda u} du = e^{-\lambda t}$.